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Palaeomagnetism of the Lizard Complex, SW England

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SUMMARY: The Lizard Complex of SW England has recently been interpreted as representing a fragment of oceanic lithosphere that was thrust on to the continental crust of S Cornwall in the Devonian as a result of tectonic processes at a convergent plate margin. A palaeomagnetic study has been carried out on the ultramafics, gabbros and dykes of the E coast section to examine possible tectonic movements that might have been associated with this history of formation. A component of stable magnetization having a shallow to intermediate negative inclination and a SSW declination has been identified in each of these lithologies. This direction is similar to the Permian geomagnetic field direction for SW England, and one interpretation of the results is that the Lizard rocks were remagnetized in the early Permian, perhaps as a result of heating associated with emplacement of the Variscan batholiths. However, no mineralogical or other evidence for widespread heating of the Lizard Complex in the Permian has yet been identified. Available Rb-Sr and K-Ar geochronological data indicate a probable Devonian crystallization age, with no evidence for a Permian overprint. Therefore it is considered equally likely that this magnetization is primary, and dates from the time of original formation of the rocks. After application of a 90° tilt correction to restore the E Coast section of the complex to its postulated original orientation in the oceanic lithosphere the corresponding palaeomagnetic pole position is brought into close coincidence with the mean UK pole for the interval Silurian to Devonian. If the stable magnetization is primary, and was acquired during the original formation of the complex by sea floor spreading at a mid-ocean ridge, then it may be inferred that movement in a latitudinal sense prior to emplacement was small, and below the resolution of the present palaeomagnetic data (10° to 15°).

The origin of the small yet remarkably interesting assemblage of mafic and ultramafic rocks that form the Lizard peninsula of S Cornwall has been the subject of extensive debate over the past century. The formerly held view that the Lizard Complex represents a high temperature intrusion of upper mantle material surrounded by a contact metamorphic aureole (e.g. Flett & Hill 1912; Green 1964) has now been largely superseded by the alternative interpretation that the ultramafic and mafic rocks were emplaced by thrusting (e.g. Hendricks 1937, 1939; Sanders 1955).

A major step forward in providing a mechanism for such large-scale thrusting was the demonstration by Thayer (1967) of the close resemblance of the Lizard assemblage to the characteristic sequences of rocks which are found in ophiolite complexes and believed to represent fragments of oceanic crust and upper mantle that have been tectonically emplaced on to the continents. More recent geological, petrological, geochemical and geophysical studies have confirmed these close similarities (e.g. Mitchell 1974; Strong *et al.* 1975; Badham & Kirby 1976; Kirby 1979).

The exact mechanism of emplacement of the Lizard ophiolite is still poorly understood, but most models involve a component of convergent motion in the mid-Palaeozoic between major plates incorporating parts of present day Africa and W Europe, associated with the closure of an intervening ocean basin of uncertain dimensions. According to this view the characteristic serpentinite-gabbro sheeted dyke assemblage of the Lizard represents a segment of oceanic crust and upper mantle which was originally formed at a mid-ocean ridge located somewhere to the S and E, and emplaced by obduction on to the continental crust of S Cornwall in the Devonian. In an attempt to test, and possibly refine these models, a palaeomagnetic study has been carried out on the rocks of the Lizard Complex. Such a study has the potential to detect any significant latitudinal motion that might have resulted from sea floor spreading prior to obduction, and possibly also to elucidate the tectonic movements associated with emplacement of the complex.

Geological summary and sampling procedures

It is believed that the E coast section of the Lizard Complex forms a distinct tectonic unit which is separated from the rest of the complex by a major thrust fault (Kirby 1979). This unit comprises a succession, from N to S, of four divisions: (i) a basic dyke complex in gabbro; (ii) massive, variable and layered gabbro; (iii) troctolites and related rocks in peridotite; and (iv) peridotite (Fig. 1). The peridotite has undergone varying amounts of serpentinization. The remainder of the Lizard Complex contains very little gabbro and is made up predominantly of peridotite together with the hornblende schists and Old Lizard Head rocks, which are believed to represent ophiolitic and sub-ophiolitic rocks that were deformed by the overriding major thrust sheets. In this study sampling was restricted to the basic dykes, gabbro, troctolites and serpentinite that comprise the

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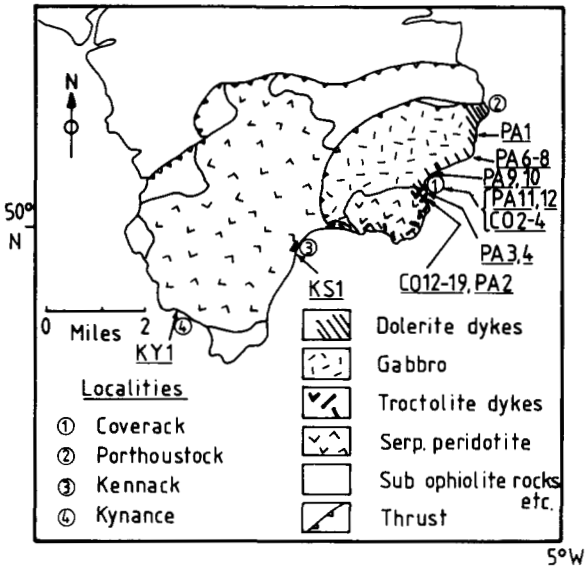


FIG. 1. Outline geological map of the Lizard Complex, showing locations of palaeomagnetic sampling sites.

main E coast section between Porthoustock and Coverack, together with the peridotite exposed at Kennack Sands and Kynance Cove (Fig. 1).

At approximately half of the sites, samples were drilled in the field using a portable two-stroke drill, which produced 2.5 cm diameter cores up to 10 cm in length. These were orientated by both magnetic and sun compass. A comparison between the two sets of readings showed that the magnetic bearings had been influenced by the nearby highly magnetic rock, leading to errors that often exceeded 20°. Consequently only the sun compass bearings were used in the palaeomagnetic analysis. At the remaining sites hand samples were collected and these were also orientated by sun compass. Up to three separate 2.5 cm right cylindrical specimens were cut from each hand sample for palaeomagnetic analysis. In the majority of cases this sampling procedure resulted in a total of between three and six separately orientated samples for each site.

Palaeomagnetic measurements

Approximately half of the remanence measurements were carried out on a Digico computerized slow-speed magnetometer (Molyneux 1971), and the rest on a microprocessor-controlled parastatic magnetometer. Between one and three separate specimens from each site were subjected to incremental alternating field (a.f.) demagnetization at various steps up to a maximum applied field of 60 mT. This treatment has the effect of preferentially removing low stability components of magnetic remanence, and its purpose was to explore the stability characteristics of the magnetization, and to isolate any stable component present. The results are presented in two forms, firstly as stereographic projections indicating the change in direction of magnetization during treatment, with accompanying graphs showing the corresponding changes in intensity, and secondly as 'vector plots' (Zijderveld 1964). The latter comprise a rectangular axis system in which the end-point of each vector is projected orthogonally on to the horizontal plane (crosses) and on to the N-S vertical plane (squares). These plots have the useful property that when only a single stable component of magnetism remains in the sample this can be readily recognized by the vectors decreasing along a straight line directed towards the origin. For those sites at which the trial samples indicated the presence of a well-defined stable-end-point the remaining samples were demagnetized at the optimum peak alternating field value deduced from the trial sample behaviour.

Results

(a) *Metadolerite*. The demagnetization behaviour of samples from two separate dolerite dykes forming part of the NW–SE trending swarm N of Coverack are shown in Fig. 2. The vector plot for sample CO17.3 shows an initial movement away from the origin during the first four demagnetization steps and then an abrupt change in trend, followed by simple movement along a straight line path towards the origin. This indicates the presence of a weak low stability component (which is removed by demagnetization at an applied field of 20 mT), superimposed on a single higher stability component. It can be seen from the stereographic projection that the latter component is characterized by a SW declination and a negative (upward-directed) inclination. The presence of a multi-component magnetization is indicated in sample PA4.2B by the steady directional trend observed in the stereographic projection up to the maximum applied field of 30 mT. Although this trend appears to be towards a similar negative SW direction to that identified in sample CO17.3, a demagnetization stable-end-point is not achieved by this treatment (as confirmed by the fact that the corresponding paths on the vector plots are not oriented directly towards the origin in Fig. 2*d*).

(b) *Gabbro*. Examples of the demagnetization behaviour of samples of gabbro from two sites located some 2 km apart on the E coast section are shown in Fig. 3. The magnetization of sample PA11.1 appears to be dominated by a high stability component having a negative SSW direction. Small-scale erratic changes occur during treatment up to 50 mT, possibly due to instrumental effects, but there is no clearly defined directional trend that might indicate the presence of more than one component of remanence. In contrast, sample PA8.1B does show a systematic trend, consistent with the removal of a low stability component during treatment up to about 20 mT (Fig. 3*c*), after

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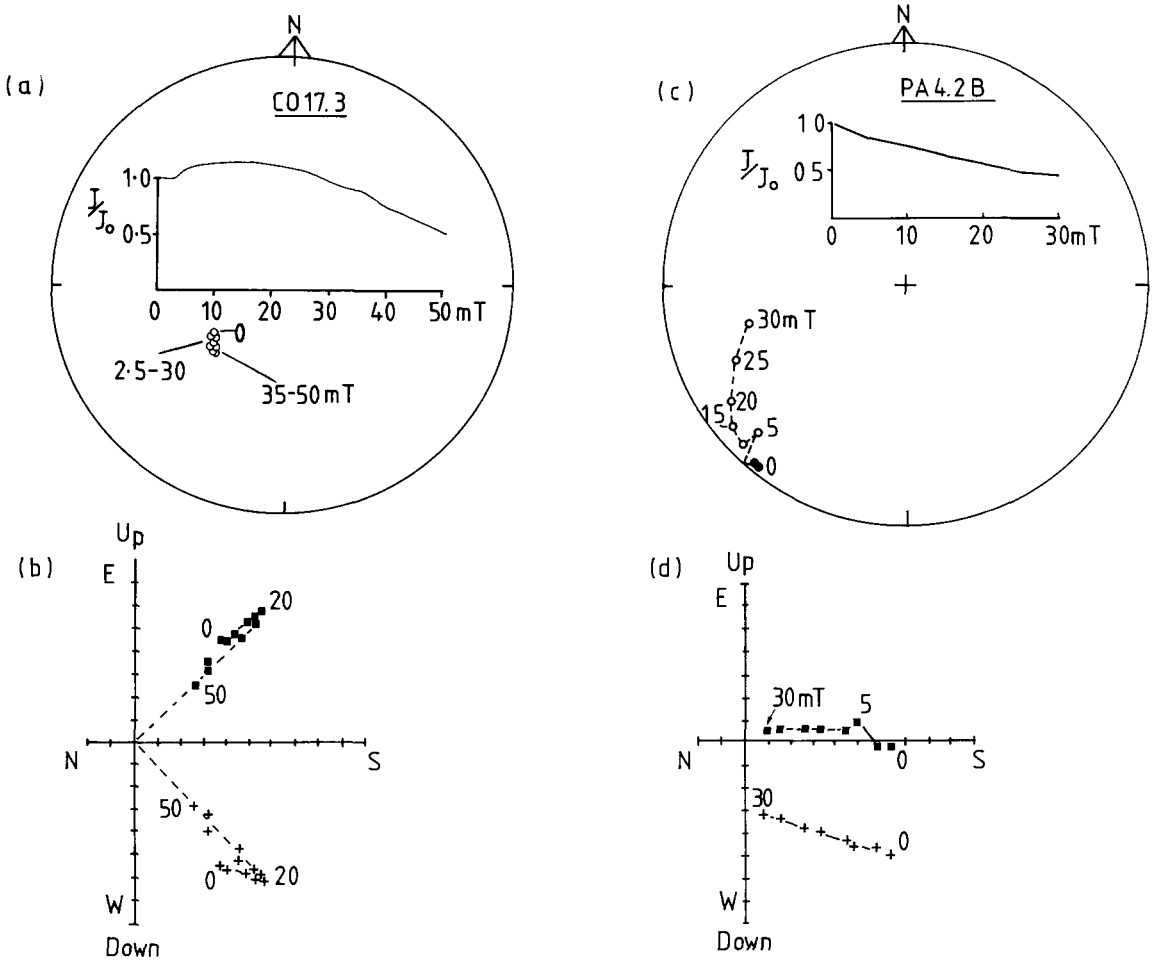


FIG. 2. Examples of response to a.f. demagnetization of typical dolerite specimens.

(*a*), (*c*) Stereographic (equal area) projections of remanence directions after successive demagnetization steps. Solid (open) symbols represent points in lower (upper) hemisphere. Normalized intensity decay curves also shown.

(*b*), (*d*) Orthogonal vector plots of same data as in (*a*) and (*c*). See text for explanation. Crosses (squares) represent points plotted on to the horizontal (vertical) plane.

which a progressive movement towards the origin (Fig. 3*d*) indicates that a stable end point has been achieved by the demagnetization process. This stable component has a SW declination, but a rather shallower inclination than that of sample PA11.1.

(c) *Troctolite.* An upward-directed SW stable component also appears to be characteristic of the troctolite dykes, as indicated by the behaviour of the two typical samples shown in Fig. 4. In both cases the removal of a less stable component by treatment at 15–30 mT is followed by a steady trend towards the origin on the vector plot. Minor fluctuations in this trend probably reflect spurious instrumental effects, which begin to become significant at applied fields above 30 mT for the particular a.f. demagnetizer used in this study.

(d) *Peridotite.* A stable magnetization having a very similar direction is present in the majority of the samples of serpentinized peridotite examined. An example of a sample showing the typical upward SSW stable direction is shown in Fig. 5 (*a* & *b*). A few samples show stable magnetizations which depart significantly from this typical direction, and an exam-

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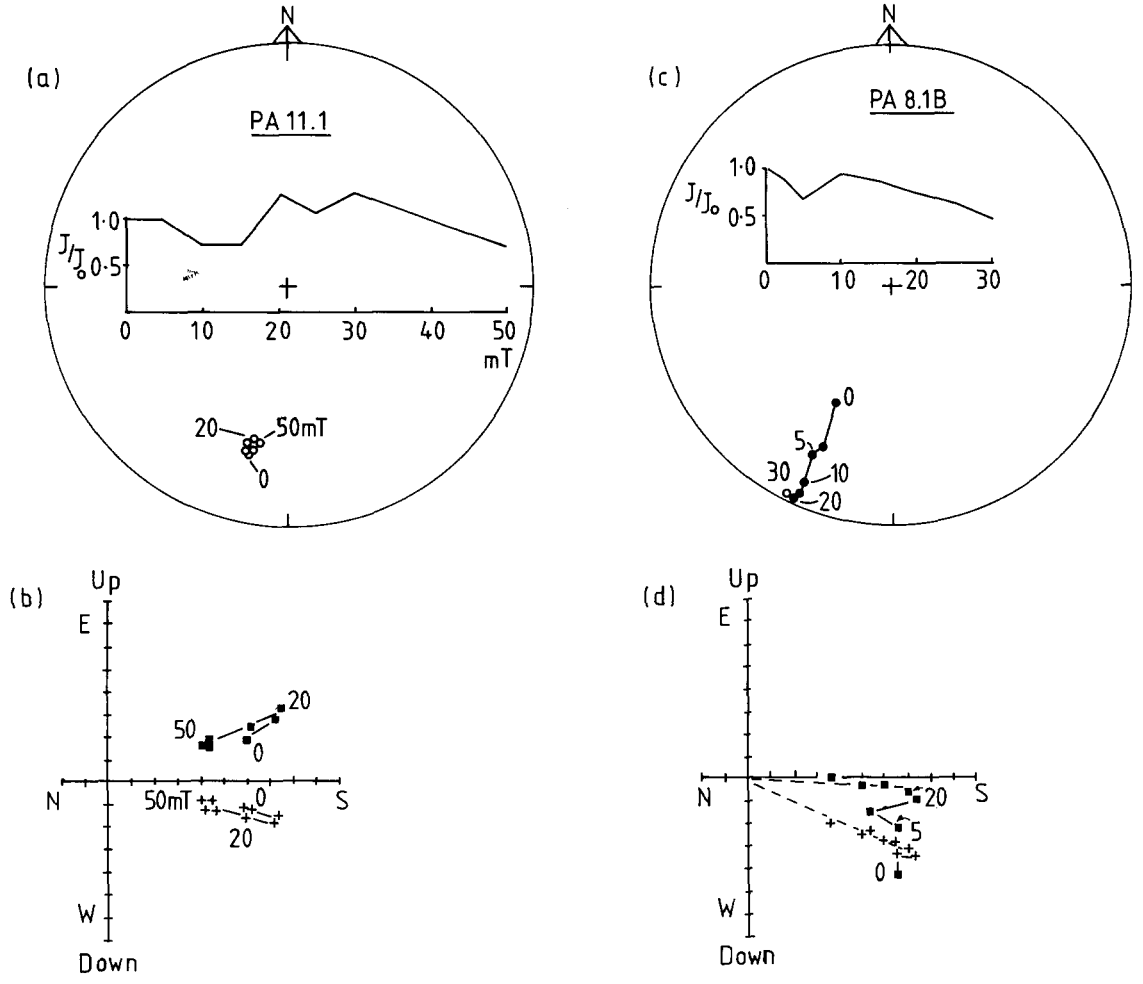


FIG. 3. Examples of response to a.f. demagnetization of typical gabbro specimens. Symbols as in Fig. 2.

ple of one with a well-defined downward (positive) SW stable direction is shown in Fig. 5 (*c* & *d*).

The examples of demagnetization behaviour discussed above indicate that each of the four lithological units studied possess a characteristic stable component of magnetization which has a shallow to intermediate negative inclination and a SSW declination. Where well-defined directional trends occur at relatively low applied fields (e.g. for samples PA8.1B and KS1.2B) these can be approximated by segments of great circles passing through the present earth's field direction at the sampling site. This suggests that the low stability component removed from these samples is probably a viscous magnetization acquired in the present field. The mean site directions of stable magnetization for all 28 sites studied are shown in Fig. 6*a*. There is a clear grouping of these directions in the negative SW quadrant with a few points lying in the negative NW quadrant and a few in the positive hemisphere. In view of the demagnetization behaviour discussed above, the most likely explanation for the spread of points into the positive hemisphere is the incomplete removal of a secondary component parallel to the present geomagnetic field. The spread of directions into the negative NW quadrant is less readily explained, but local tectonic rotations cannot be completely ruled out. All four sites with directions in this quadrant are located close to the gabbro-peridotite contact at Coverack, but other sites in the immediate vicinity possess typical negative SW magnetization.

There is a relatively clear cut-off betwen those sites which are characterized by the 'typical' negative SW direction and those with the more scattered 'atypical'

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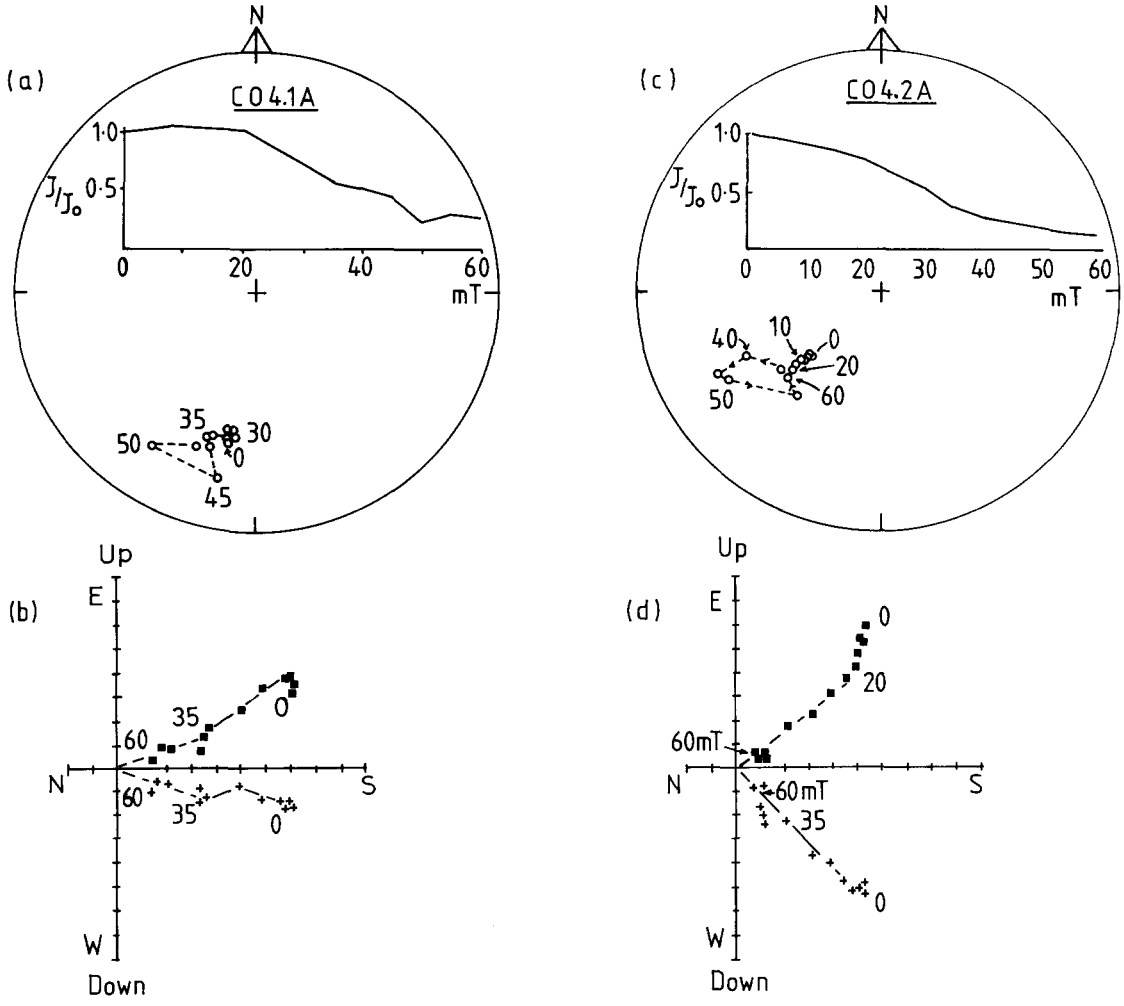


FIG. 4. Examples of response to a.f. demagnetization of typical troctolite specimens. Symbols as in Fig. 2.

directions. The former are shown separated from the latter in Fig. 6*b* and the overall mean directions of magnetization for the points plotted in these two diagrams are listed in Table 1. Mean site directions from all four lithologies contribute to the cluster of 'typical' directions, whereas 'atypical' directions appear to be restricted to samples of serpentinite and gabbro.

Opaque mineralogy

In order to assess whether the occurrence of the 'atypical' magnetization vectors is mineralogically controlled, polished sections of seven specimens showing 'typical' stable remanence vectors and five showing 'atypical' vectors were subjected to detailed microscopic examination. The majority of these samples were serpentinized peridotites.

All specimens show skeletal magnetite as a late-crystallization feature. During serpentinization magnetite was developed along the hydrating fractures in fine blebs and veinlets. Some sulphide (mostly pyrite, but some chalcopyrite) was also formed at this time. The early skeletal magnetite appears to have been slightly fractured, but otherwise unaffected by this process. In most samples, progressive mineralogical changes due to later oxidation can be identified. Firstly, the serpentine itself has been oxidized, the resulting haematite causing the pervasive redness of the samples. Secondly, the syn-serpentinization magnetite has been progressively replaced by haematite (as also has the sulphides). Finally, the early skeletal magnetites

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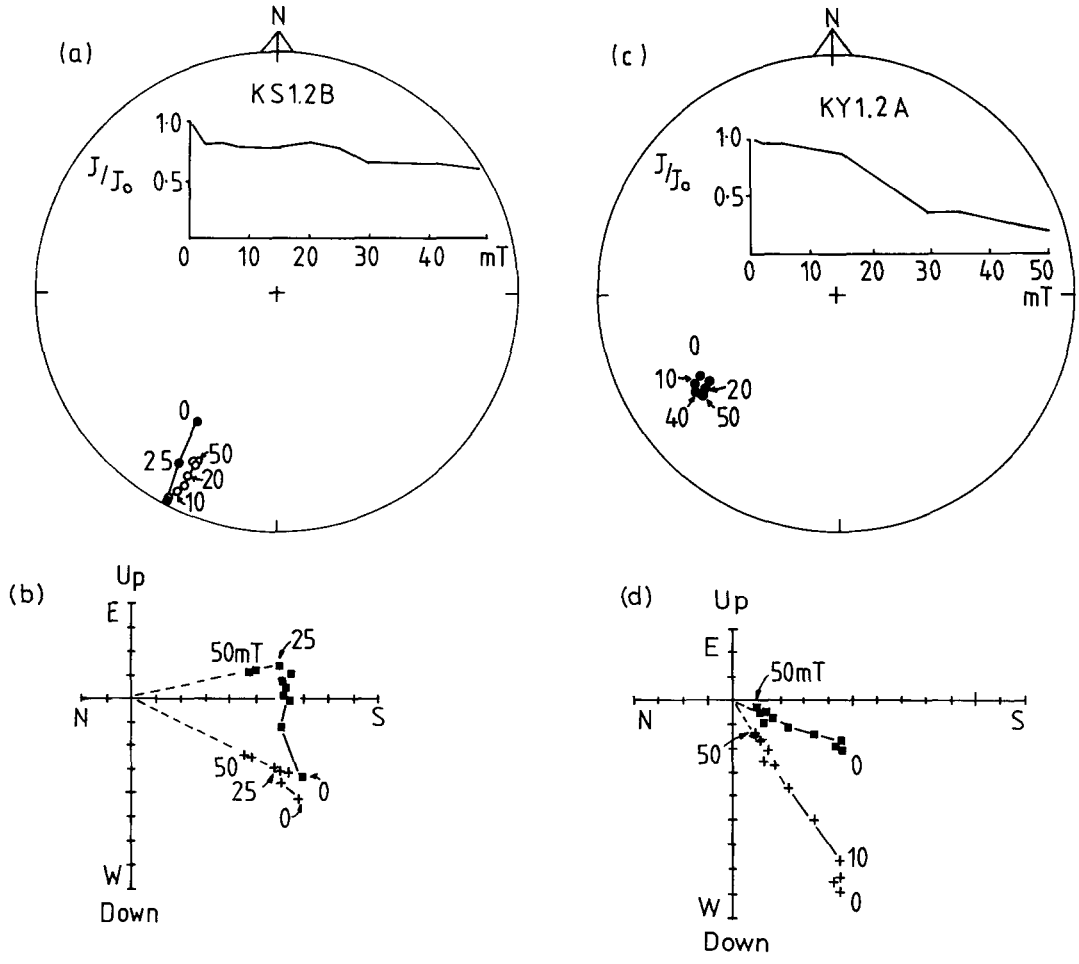


FIG. 5. Examples of response to a.f. demagnetization of typical serpentinite specimens. Symbols as in Fig. 2.

have been progressively replaced (to a varying degree) by haematite, first along rims and fractures, and then more pervasively. A lack of rutile and ilmenite in the replaced magnetites indicates a primary low-Ti nature. The main mineralogical distinction between the 'typical' and 'atypical' samples is that the former appear to show no oxidation of primary skeletal magnetite, whereas the latter show partial or complete oxidation of these grains.

A possible interpretation of the palaeomagnetic data, therefore, is that stable palaeomagnetic remanence of the 'typical' samples is carried by primary or syn-serpentinization magnetite, whereas that of the 'atypical' samples is carried by relict syn-serpentinization magnetite and/or late haematite, and therefore was acquired at a later date. Unfortunately, on the basis of the available data, it is not possible to establish the timing of the successive alteration processes with respect to possible on-ridge or obduction events.

Discussion

Convincing arguments have been put forward by Kirby (1979) that the uppermost sheet of the Lizard Complex (from which most of the samples in this study were obtained) was turned on to its side during emplacement. On the basis of evidence discussed by Kirby it is reasonable to assume that this tilting involved a rotation through an angle close to 90° about an E–W axis. If the principal magnetization of the Lizard rocks pre-dates the tilting and was acquired when the complex was first formed at a mid-ocean ridge, then a 90° tilt correction should be applied to the mean direction of magnetization before attempting to compare it with that for rocks of similar age elsewhere in western Europe.

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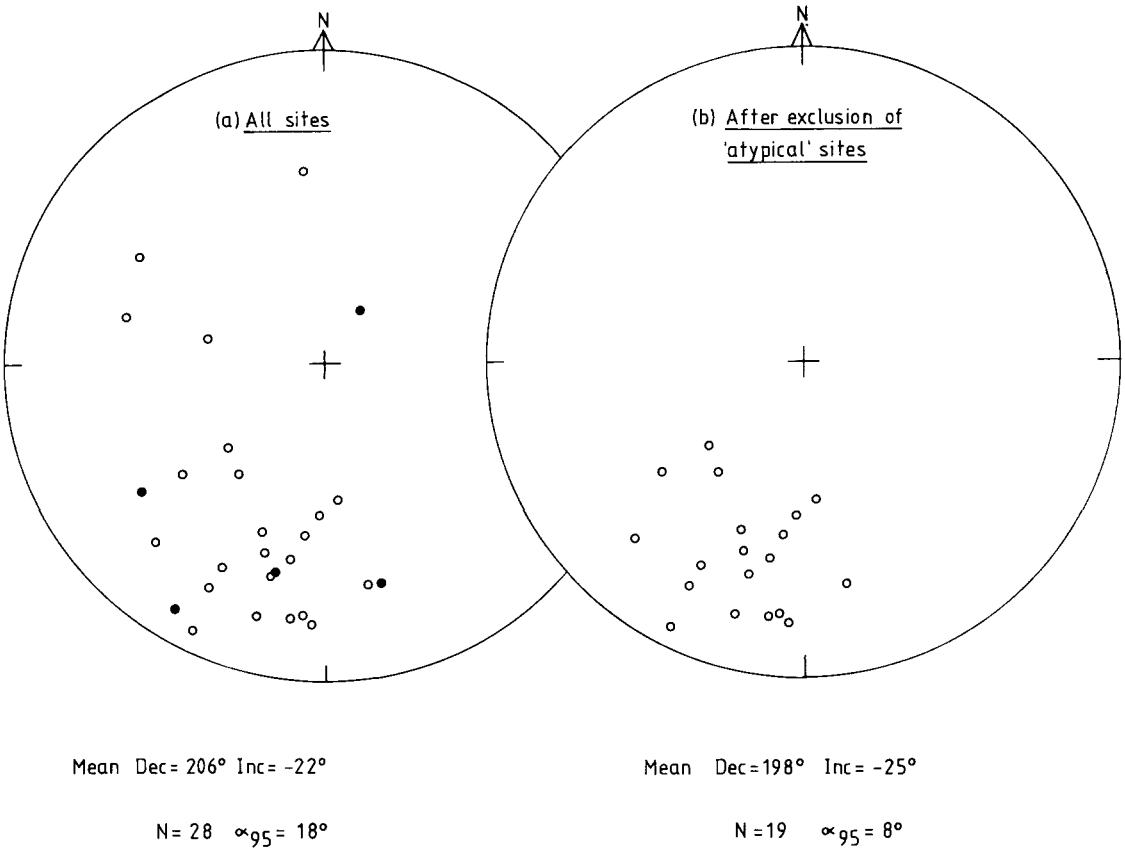


FIG. 6. Mean site directions of stable magnetization. Symbols as in Fig. 2*a*.

Radiometric age determinations indicate a probable early Devonian age for the Lizard Complex (e.g. Halliday & Mitchell 1977). The palaeomagnetic pole position calculated from the tilt-corrected mean direction of stable magnetization is therefore compared with the mean Silurian–Devonian pole for the UK and Scandinavia (i.e. sites on the Euramerican plate immediately N of the Lizard) in Fig. 7*a*. Within the 95% confidence circles there is no significant difference between these two poles. This agreement is quite striking, especially considering the uncertainty (perhaps ~10°) in the value of the 90° tilt-correction applied. If the magnetization of the Lizard rocks is primary, and was acquired during their initial formation at a mid-ocean ridge, then these results indicate that the total amount of latitudinal motion of the Lizard prior to emplacement was small and certainly less than the limits of detection of the palaeomagnetic data (10–15°).

However, an alternative interpretation of these new palaeomagnetic data is also possible, involving remagnetization of the Lizard Complex during the Permian. The Ordovician–Triassic apparent polar wander path for western Europe (from Duff 1980) is plotted in Fig. 7*b*, together with the palaeomagnetic pole position for the Lizard Complex calculated from the mean direction of stable magnetization *before* application of the tilt correction. Within the 95% confidence limits this pole is not significantly different from the mean Permian pole for western Europe, and this raises the possibility of Permian remagnetization of the Lizard Complex. A major episode of igneous activity represented by the Variscan granite belt occurred in SW England close to the Carboniferous–Permian boundary. The nearest granite batholith to the Lizard Complex is the Carnmenellis granite, but the thermal aureole of this mass does not extend S of the Helford river. Therefore it is improbable that the magnetization of the Lizard rocks could have been re-set as a result of direct heating by the granite. Furthermore,

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TABLE 1. *Mean site directions of stable magnetization (before tilt correction)*

Site number	*Rock type*	*Dec*	*Inc*	*N*	α_{95}†		
KS1.1	Serpentinite	188	−12	1	—		
KS1.2	Serpentinite	207	−13	2	—		
KY1.1	Serpentinite	206	−19	3	8		
*KY1.2	Serpentinite	234	+20	1	—		
CO2.1	Serpentinite	197	−26	2	—		
CO3.1	Troctolite	190	−26	2	—		
CO4.1	Troctolite	186	−32	3	9		
CO4.2	Troctolite	231	−31	3	10		
*CO12.1	Serpentinite	282	−50	1	—		
*CO12.2	Serpentinite	355	−28	1	—		
CO14.1	Serpentinite	175	−42	3	8		
CO15.1	Serpentinite	182	−38	2	—		
CO15.2	Serpentinite	216	−42	2	—		
*CO16.1	Serpentinite	284	−26	1	—		
CO17.3	Dolerite	228	−46	1	—		
*CO18.1	Serpentinite	35	+68	1	—		
CO19.1	Troctolite	169	−19	2	—		
*PA1	Gabbro	211	+6	6	25		
PA2	Serpentinite	200	−31	6	6		
PA4	Dolerite	206	−4	6	17		
PA5	Gabbro	183	−11	5	31		
*PA6	Gabbro	166	+19	5	36		
PA7	Dolerite	185	−13	6	10		
PA8	Gabbro	195	−11	6	13		
*PA9	Gabbro	301	−23	6	24		
PA10	Gabbro	223	−15	4	29		
PA11	Gabbro	194	−21	2	13		
*PA12	Gabbro	193	+22	2	18		
						Palaeomagnetic South Pole position	
Overall mean						*Lat.*	*Long.*
(i) All sites		206	−22	28	17	46S	316E
(ii) After exclusion of 'atypical' site directions (†) in above table		198	−25	19	8	50S	327E
(iii) After application of tilt correction to (ii)		213	+60	19	8	5N	330E

† Radius of 95% confidence circle about mean direction (calculated for sites with $N > 2$)

petrologic observations have revealed no evidence of changes in the magnetic mineralogy of these rocks that might be associated with Hercynian events, and published geochronological data reveal no evidence for a radiometric re-setting of the rocks in the Permian. Although the possibility of Permian remagnetization cannot be completely ruled out, we tend to favour the alternative interpretation that the principal stable magnetization of the Lizard rocks was acquired during their initial formation in an ocean floor environment and that this magnetization has survived the tectonic processes associated with emplacement of the ophiolite. This interpretation implies that rotation about vertical axes during emplacement was unimportant compared with the major northward overturning about an E–W axis. Furthermore, the agreement of the stable magnetization directions for three out of the four sites situated at Kennack Sands and Kynance Cove with those for the E coast section suggests that the separate thrust sheet in which these sites are situated was also involved in the northward overturning.

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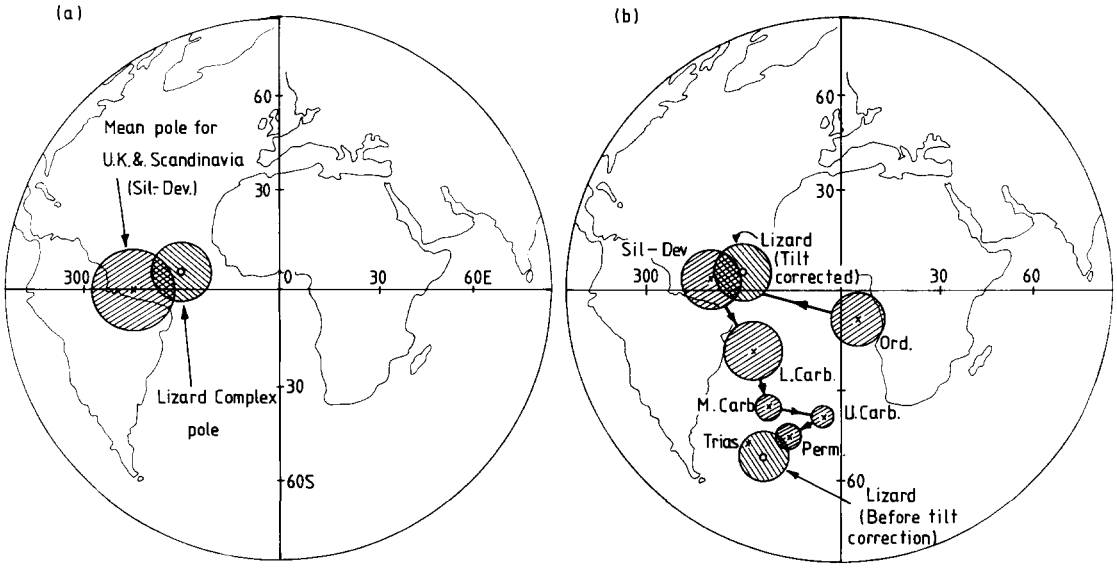


FIG. 7. (*a*) Comparison between pole position for Lizard complex (after exclusion of 'atypical' data and application of tilt correction) and mean palaeomagnetic pole position for UK and Scandinavian sites. (*b*) Comparison of Lizard palaeomagnetic pole positions before and after application of tilt correction with Ordovician–Triassic apparent polar wander path for western Europe (Duff 1980).

References

BADHAM, J. P. N. & KIRBY, G. A. 1976. Ophiolites and the generation of ocean crust: data from the Lizard Complex, Cornwall. *Bull. geol. Soc. Fr.* **7,** 885–8.

DUFF, B. A. 1980. The palaeomagnetism of Jersey volcanics and dykes, and the Lower Palaeozoic apparent polar wander path for Europe. *Geophys. J. R. astron. Soc.* **60,** 355–75.

FLETT, J. S. & HILL, J. B. 1912. Geology of the Lizard and Meneage (explanation of Sheet 359). *Mem. geol. Surv. G.B.* 2nd. ed., 1946.

GREEN, D. H. 1964. A re-study and a re-interpretation of the geology of the Lizard peninsula, Cornwall. *In:* HOSKING, K. F. G. & SHRIMPTON, E. J. (eds.). *Present views on some aspects of the geology of Devon and Cornwall.* R. geol. Soc. Cornwall, Penzance, 87–114.

HALLIDAY, A. N. & MITCHELL, J. G. 1977. K-Ar ages of basaltic dykes from the Lizard Complex. *Proc. Ussher Soc.* **4,** 24.

HENDRICKS, E. M. L. 1937. Rock succession and structure in south Cornwall, a revision. *Q. J. geol. Soc. London,* **93,** 322–67.

—— 1939. The Start–Dodman–Lizard boundary zone in relation to the Alpine structure of Cornwall. *Geol. Mag.* **76,** 385–402.

KIRBY, G. A. 1979. The Lizard complex as an ophiolite. *Nature, London,* **282,** 58–61.

MITCHELL, A. H. G. 1974. Southwest England granites: magmatism and tin mineralisation in a post-tectonic setting. *Trans. Inst. Min. Metall.* Sec. B, **83,** 895–7.

MOLYNEUX, L. 1971. A complete results magnetometer for measuring the remanent magnetisation of rocks. *Geophys. J. R. astron. Soc.* **24,** 429–33.

SANDERS, L. D. 1955. Structural observations on the south-east Lizard. *Geol. Mag.* **92,** 231–40.

STRONG, D. F., STEVENS, R. K., MALPAS, J. G. & BADHAM, J. P. N. 1975. A new tale for the Lizard. *Proc. Ussher Soc.* **3,** 265.

THAYER, T. P. 1967. Chemical and structural relations of ultramafic and feldspathic rocks in alpine intrusive complexes. *In:* WYLLIE, P. J. (ed.) *Ultramafic and Related Rocks.* Wiley, New York. 222–39.

ZIJDERVELD, J. D. A. 1964. A.C. demagnetisation of rocks: analysis of results, *In:* COLLINSON, D. W., CREER, K. M. & RUNCORN, S. K. (eds.). *Methods in Palaeomagnetism.* Elsevier Pub. Co., 254–86.

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